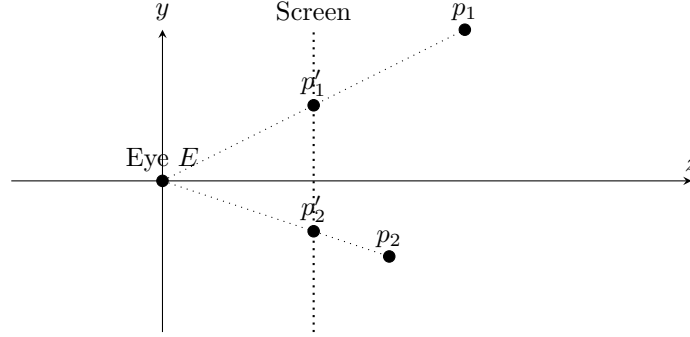


1 3D Projection



$$p = (p_x, p_y, p_z) \quad (1)$$

$$E = (E_x, E_y, E_z) \quad (2)$$

$$S : z = S_z \quad (3)$$

The formula to find (p'_x, p'_y, p'_z)

$$p'_x = \frac{(p_x - E_x)(S_z - E_z)}{(p_z - E_z)} + E_x \quad (4)$$

$$p'_y = \frac{(p_y - E_y)(S_z - E_z)}{(p_z - E_z)} + E_y \quad (5)$$

$$p'_z = S_z \quad (6)$$

let $E = (0, 0, 0)$ and $S_z = 1$, then

$$p'_x = \frac{p_x}{p_z}, \quad p'_y = \frac{p_y}{p_z}, \quad p'_z = 1 \quad (7)$$